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Investigating the Defect Behavior and Electronic Properties of Formamidinium Lead Bromide Perovskite through Machine Learned Interatomic Potentials

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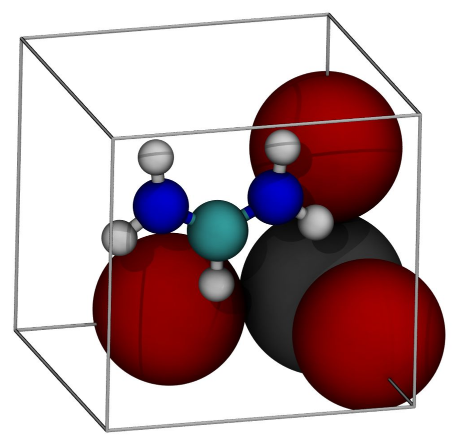
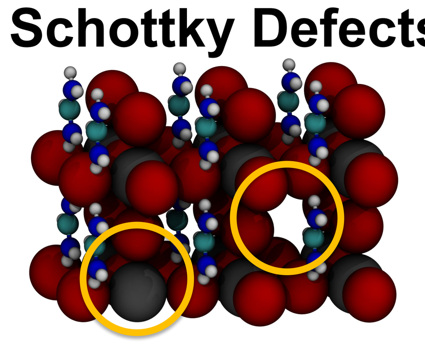
Introduction

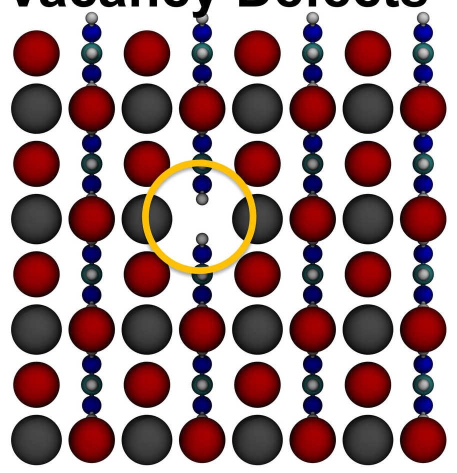


- $FAPbBr_3$ is a promising material for many optoelectronic devices, including light emitting diodes, photoelectrodes, and solar fuel cells
- Electronic properties affected by presence of defects in the crystal

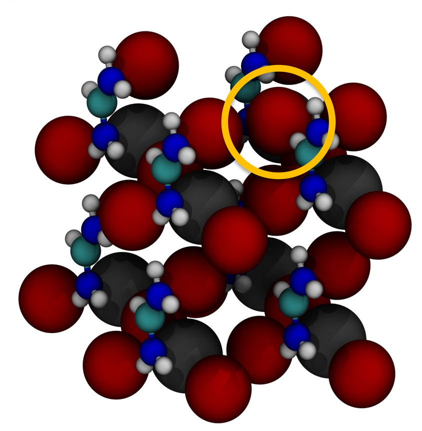
Schottky Defects



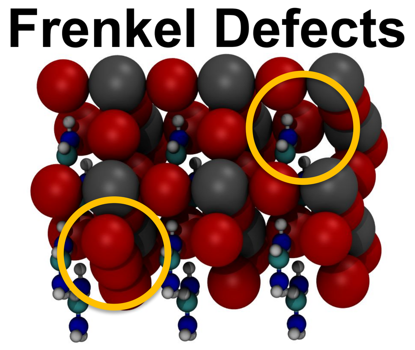
Vacancy Defects



Interstitial Defects



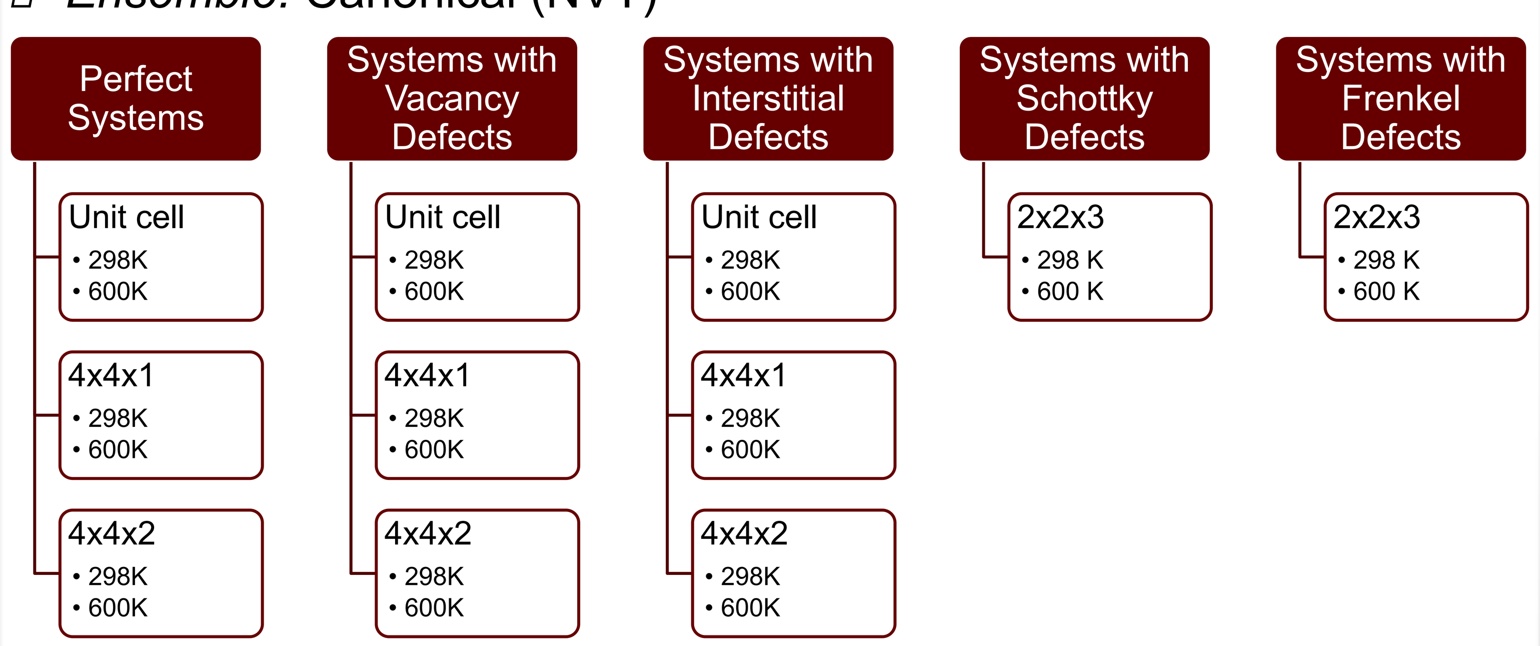
Frenkel Defects



Computational Details

Ab initio Molecular Dynamics Simulations

- *Basis set:* DZVP-MOLOPT-SR-GTH
- *Pseudopotential:* GTH-PBE
- *XC Functionals:* GGA-X-PBE, GGA-C-PBE
- *Timestep:* 0.5 fs
- *Ensemble:* Canonical (NVT)



Electronic Structure Calculations

- *DFT Functionals:* DFT-D3 for geometry optimization, HSE06 for band structure calculations
- *Gaussian Smearing:* σ = 0.05
- *Cutoff Energy:* 500 eV
- *Energy Difference:* 1×10^{-6} eV
- *Kpoint Grid:* 4 × 4 × 4 Monkhorst-Pack

Training Machine Learning Models

- *Number of Features:* 64
- *Maximum Body Order:* 3
- *Angular Momentum (l):* 1
- *Cutoff Radius:* 5 Å
- *Number of Layers:* 2

Results & Discussion

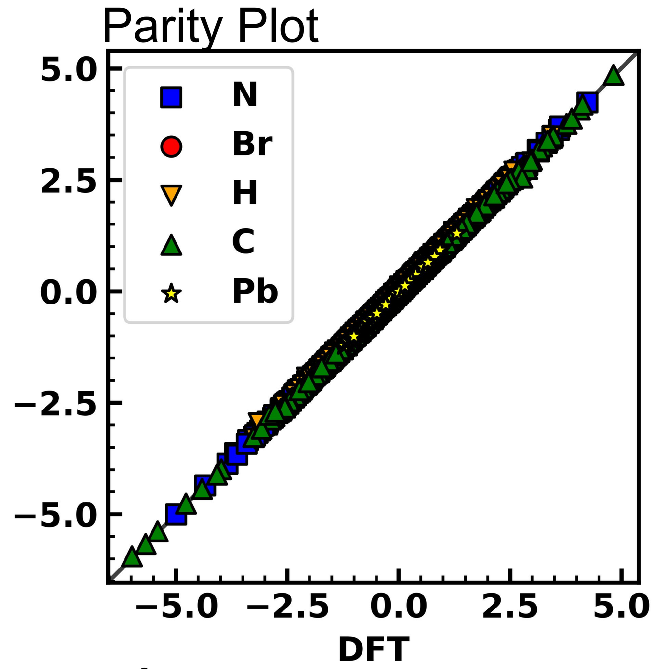
Models Trained

	No Defects (Perfect)	Defects (Frenkel and Schottky)
Bulk	Model 2	Model 3
Bulk + NPL	Model 4	Model 5

	Train			Validate		
Model	**RMSE E (meV/atom)**	**RMSE F (meV/Å)**	**Relative F RMSE (%)**	**RMSE E (meV/atom)**	**RMSE F (meV/Å)**	**Relative F RMSE (%)**
Model 0	9.3	103.3	14.53	5.3	119.7	16.55
Model 1	8.0	90.9	12.62	4.2	82.6	11.48
Model 2	0.3	8.6	1.04	0.3	12.5	1.43
Model 3	4.9	72.0	9.79	6.0	70.3	9.46
Model 4	2.2	93.2	11.98	2.5	108.0	13.23
Model 5	4.9	117.5	15.54	3.6	127.5	16.30

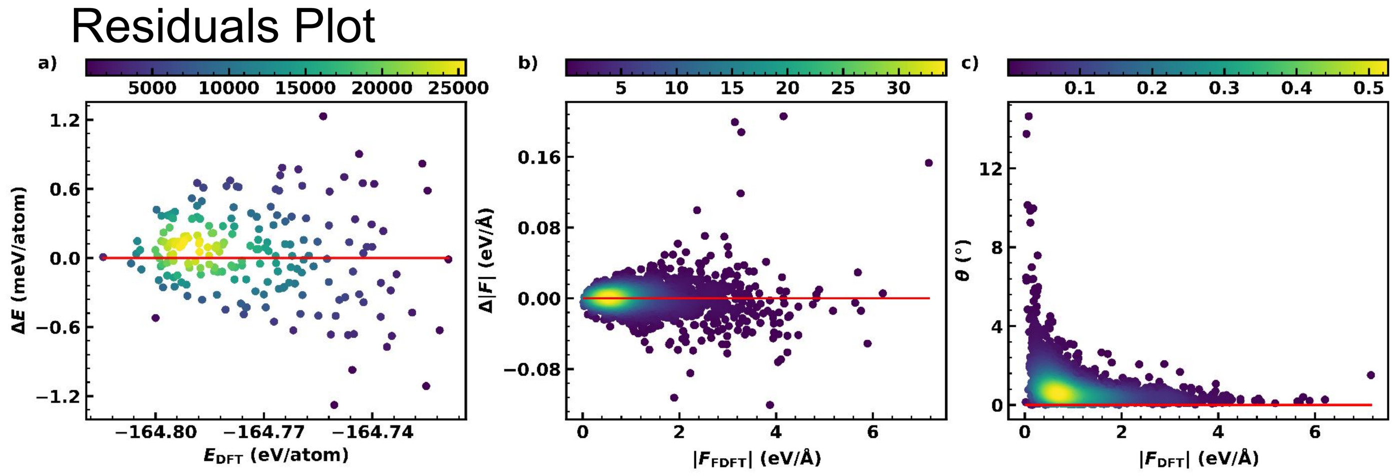
Performance of Model 2

Parity Plot

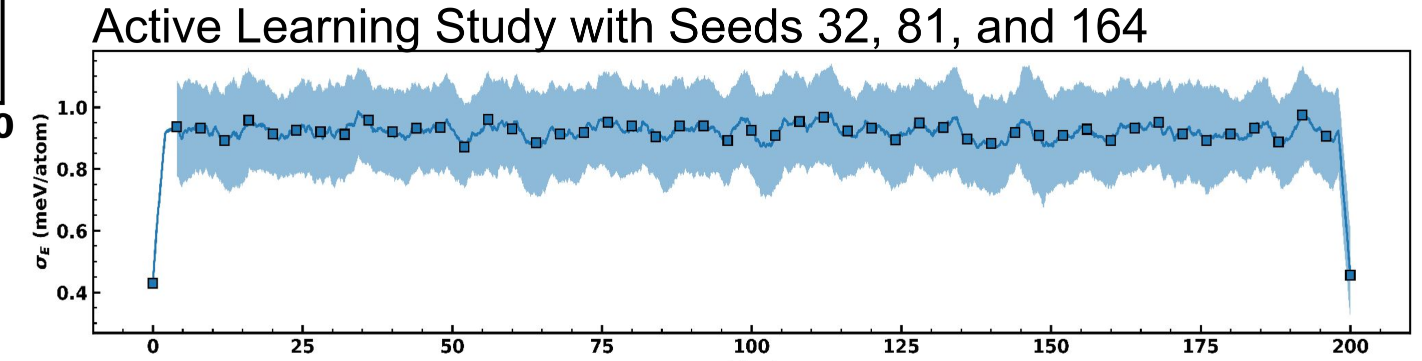


R^2 *Values:* N = 0.9998, Br = 0.9994, H = 0.9997, C = 0.9998, Pb = 0.9997

Residuals Plot

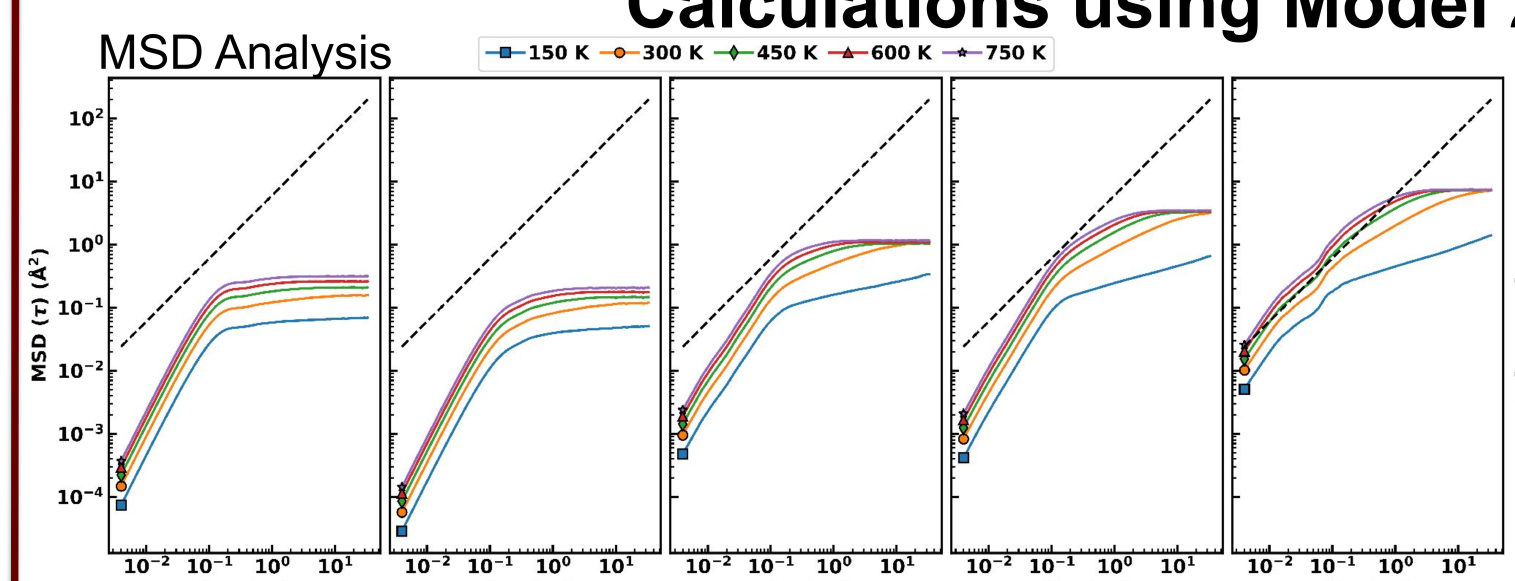


Active Learning Study with Seeds 32, 81, and 164

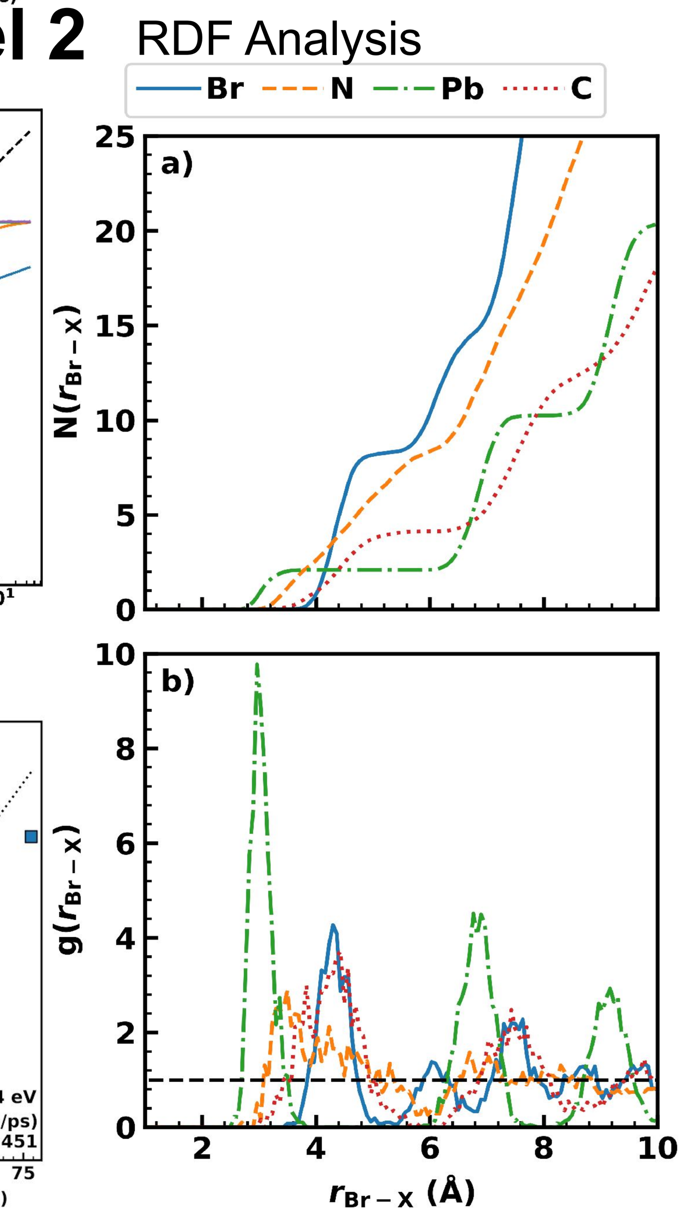


Calculations using Model 2

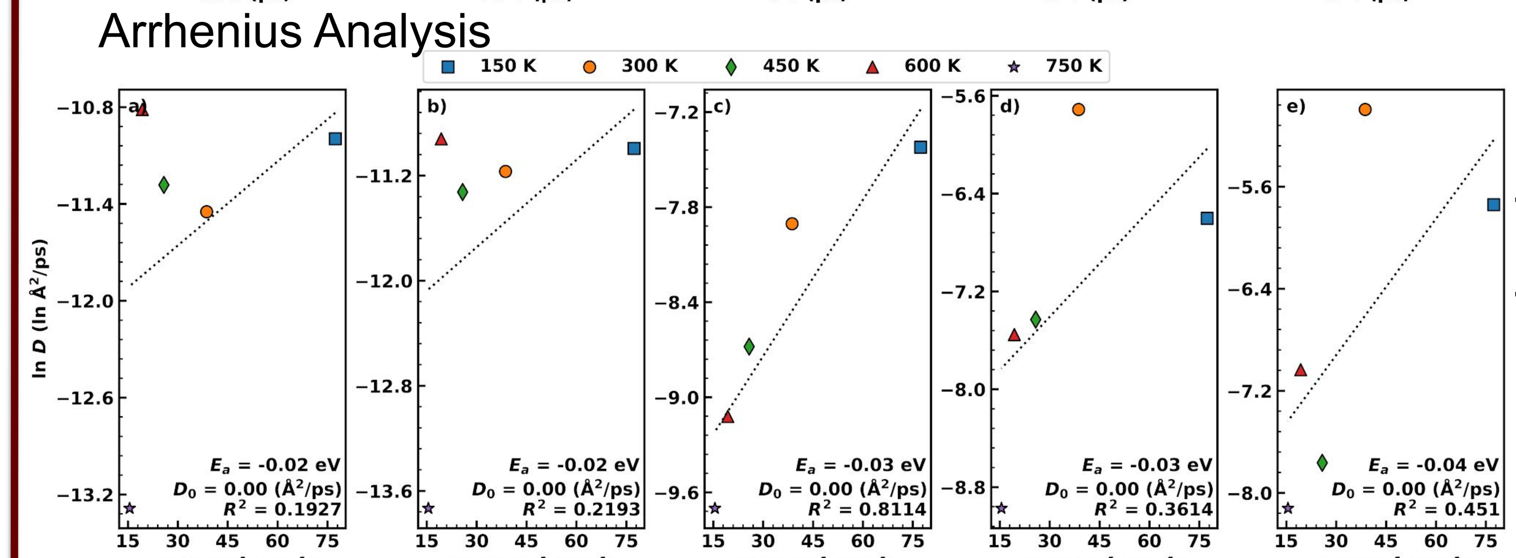
MSD Analysis



RDF Analysis

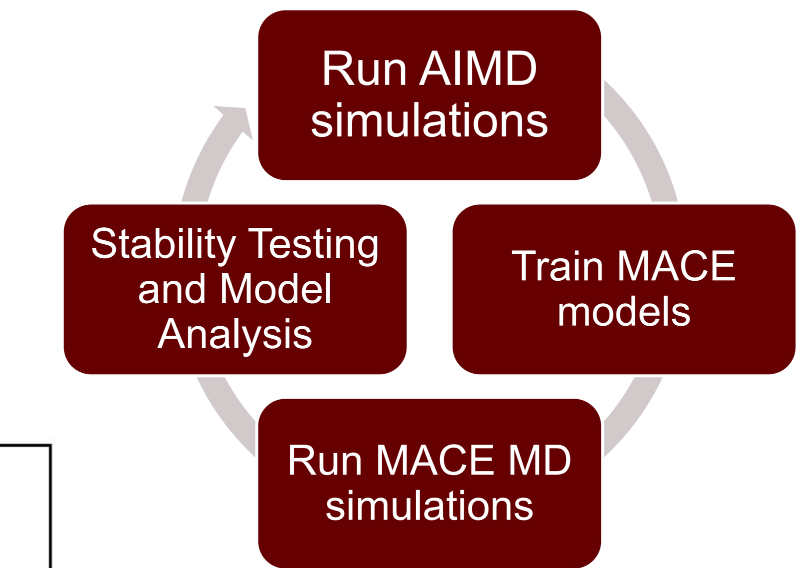


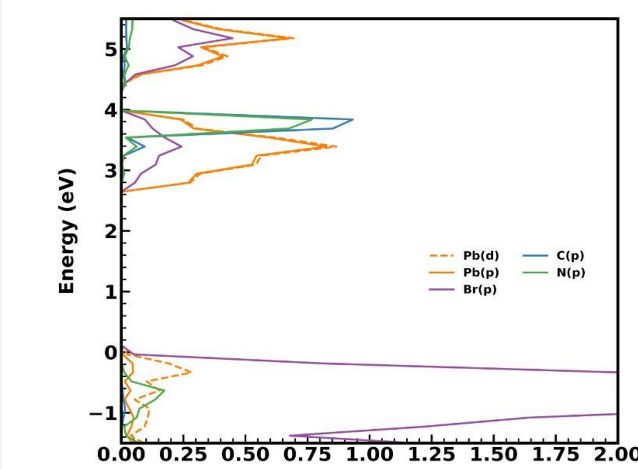
Arrhenius Analysis

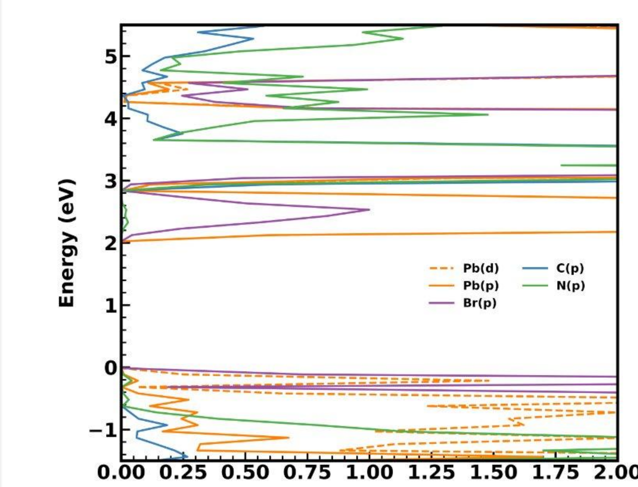


Future Work

- Continue training models to reduce model error
- Band structure and density of states calculations

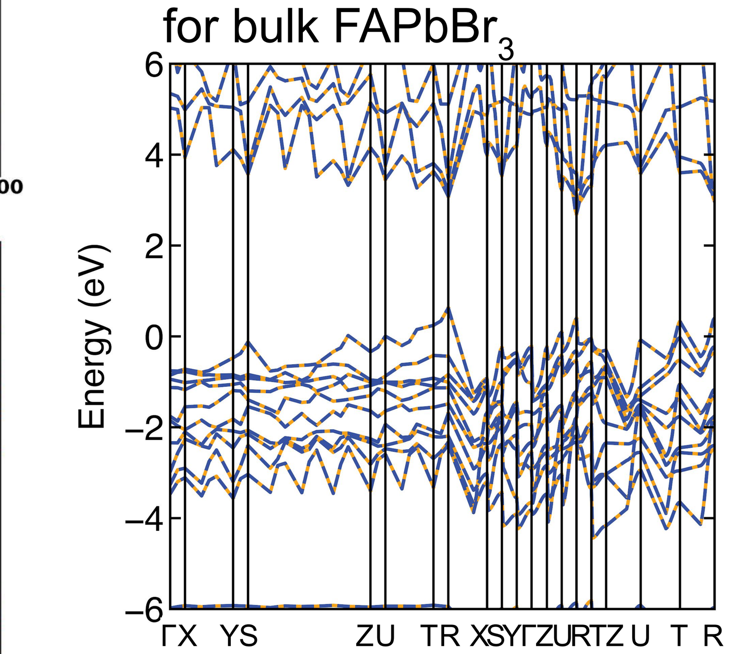






PDOS plots of bulk (top) and 4x4x1 (bottom) $FAPbBr_3$

Band structure calculation for bulk $FAPbBr_3$



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Acknowledgments

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